# Universal validity is decided on finite universal models

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openlogic:normal-modal-logic/filtrations:local-snapshot-20261013;
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*Editorial note.* The author's left-to-right sentence opens with 'trivial'; on reread it is the one-step definitional inclusion of finite universal models in the universal class and nothing more, so the word is earned.

*Editorial note.* Fix note (duty-manager report 2026-10-14, post-park card-face repair): the stub number field of external reference prop:S5=univ contained the equals sign inherited from the source label; the number field is the card-face stub naming layer, not the external label layer, so it was changed to prop-S5-univ while the label field keeps the true source label prop:S5=univ. No other field touched.

*Editorial scope.* One original proposition of the Open Logic Project unit normal-modal-logic/filtrations, counted once: universal validity collapses to finite universal validity. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

*Allowed standard prerequisites (editorial).*

- Universal models and their finite counterparts: the class U of universal models (every pair of worlds related) and its finite subclass U-Fin, per the frame-definability unit's characterization cited in the statement.
- The filtration definition of this unit with its three conditions (defn:filtration and its R-condition, in particular R1: related worlds in M give related classes in M-star) and the finiteness bound prop:filt-are-finite.
- The filtration truth lemma (thm:filtrations) transferring satisfaction of formulas in Gamma between a model and its filtration.

*Source reference.* Original proposition prop-S5-univ: sources/olp-normal-modal-logic-frame-definability/pinned-original-source.tex, source lines 657–657. The statement cites the frame-definability unit's characterization of universal models for the class U; the defining unit is a sibling OLP unit, so the reference records the label in that unit's pinned source.

*Source reference.* Original proposition prop-filt-are-finite: sources/olp-normal-modal-logic-filtrations/pinned-original-source.tex, source lines 1103–1103. The

author proof cites the class-count bound; this proposition is delivered as the corpus card X2-logic-modal-filtrations-finite of the same unit.

*Source reference.* Original theorem thm-filtrations: sources/olp-normal-modal-logic-filtrations/pinned-original-source.tex, source lines 943–943. The author proof cites the filtration truth lemma to keep A false at the class of w in the filtration; that lemma is parked (its case induction sits behind iftag conditionals), so the reference records its source label line.

*Source reference.* Original definition defn-filtration: sources/olp-normal-modal-logic-filtrations/pinned-original-source.tex, source lines 911–911. The author proof appeals to the filtration definition for the finiteness construction and for the universality check; the reference records the definition's label line in this unit.

*Source reference.* Original definition defn-filtration-R: sources/olp-normal-modal-logic-filtrations/pinned-original-source.tex, source lines 917–917. The universality check of the author proof uses the R-condition of the filtration definition (related worlds give related classes); the reference records that condition's label line in this unit.

**Proposition 0.1.** *Let $[U]$ be the class of universal models (see prop-S5-univ ) and $[U]_{\mathrm{Fin}}$ the class of all finite universal models. Then any formula $A$ is valid in $[U]$ if and only if it is valid in $[U]_{\mathrm{Fin}}$.*

*Proof.* Finite universal models are universal models, so the left-to-right direction is trivial. For the right-to left direction, suppose that $A$ is false at some world $w$ in a universal model $M \models$. Let $\Gamma$ contain $A$ as well as all of its subformulas; clearly $\Gamma$ is finite. Take a filtration $M^* \models$ of $M \models$; then $M^* \models$ is finite by prop-filt-are-finite , and by thm-filtrations , $A$ is false at $[w]$ in $M^* \models$. It remains to observe that $M^* \models$ is also universal: given $u$ and $v$, by hypothesis $Ruv$ and by defn-filtration defn-filtration-R , also $R^*[u][v]$. □